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Petri Net Based Business Process Simulation and Analysis Technology

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Abstract

The theoretical basis and implementation solution of Business Process Simulation and Analysis System (BPSAS) is put forward in this paper. With workflow technology, Petri Net theory and simulation technology, the workflow engine independent business process analyzing system is realized to provide Petri Net based business process analysis. Further more, an example is illustrated to demonstrate how to analyze business process in order to select optimized model. And at the same time, as a business process analyzing tool, it supports the specified BPM (Business Process Management) process.

1. Introduction

In the information economy era characterized by "Customers", "competition" and "changes", how to face the fierce market competition environment, unpredictable market risks and opportunities brought by market changes, to solve the ever-changing and rising needs of customers, is becoming the key issue of constraining survival and development of enterprises. As to the core business focused on by enterprises, the business problems like lack of effective coordination of processes, inefficient functioning of business should be avoided to adapt to rapid changes of market demand. Business Process Management (BPM) theory demands to establish a management mechanism in enterprises to manage and optimize business processes continuously.

Similarly a Chinese operation business company hopes to improve its core competence by improving IT systems for group clients business. Based on this environment, this paper determines the goal, scope and construction solution of Business Process Simulation and Analysis System (BPSAS) with the expectation of providing a set of business process analysis tools that can help to compare different business process models and provide visualization of flow charts and business process performance analysis report. On this basis, the comparison of various process models and providing of data needed by construction and optimization of business process selection can be accomplished to achieve the goal of optimizing models.

2. Background

2.1. WFMS and Improved BPM Model

Management of business processes is gradually separated to form an independent system, which is called workflow management system, with the constant evolution of IT systems. People's intuitive thinking is to separate processes from application systems.

Workflow is the calculation model for business processes [1]; it organizes processes, business logic and rules together, expresses them with appropriate models in computers and makes calculations of them. And the major function of workflow management system (WFMS) is to define, implement and manage workflows by computer-aid methods, coordinate the information interaction between tasks and group members in the course of implementation of workflow [2]. Workflow needs to be implemented by WFMS.

At present, many companies have developed their own workflow products based on these standards. There are two main categories of workflow products:

a) Fundamental workflow systems provide engines, designers and relevant interfaces. Based on these systems, developers of application systems can develop software that has functions of workflow management. Typical products are WebShpere Integration Developer (WID) and WebShpere Process Server (WPS) developed by IBM, Action Workflow from Action Technologies Inc. and open source systems such as OSWorkflow and JBPM developed by JBoss [3][4][5] [6].

b) The other category involves application-oriented software including workflow technology or internal workflow functions. These systems are end-user-oriented and often integrate other features according to

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application needs. Typical products include EasyFlow, which is a soft taking workflow technology as core and developed by the Digital China Company.

Traditional workflow system provides workflow engines and modeling tools while BPSAS founds a model analysis system based on Petri Net. This system can provide performance analysis for workflow model and offer support for the processing model optimization in the course of improving BPM.

2.2. Business Process Analysis and Evaluation

To analysis the performance of a workflow, the KPI institution of performance evaluation should be determined first. The KPI of workflow performance can be divided into two categories:

1. From the outside point of view, a specific case includes: FT (Finishing Time), WT (Waiting Time) and PT (Processing Time) of the workflow case. The relationship among them is given below:

$$FT = WT + PT$$

2. From the internal point of view, the relation between μ (Utilization of service unit) , which is used to measure the performance of service unit, CI (Case Intensity) and PC (Processing Capability) is shown as follows:

$$\mu = \frac{CI}{PC}$$

The workflow model can be optimized based on parameters above to obtain relative better performance indicators according to related principles, which provide basis for model selection. Expensive cost of re-modifying models can be avoided in the phase of practical operation of workflow.

2.3. Workflow Simulation

Workflow simulation implements various activities in the workflow model and promotes workflow cases automatically before the implementation of workflow systems. A set of original data about workflow models can be obtained by many times of simulation, and then be analyzed to get the performance data, based on which users can make analysis and evaluation about every performance during the course of operation in enterprises. Problems should be fixed and processes be redesigned to guarantee the full understand of the feasibility and efficiency of system implementation, as well as discovering the performance characteristics of business process models. Simulation of workflow cases is the basis of analysis, and simulation of system incident should be implemented because the beginning of instances and following operation occur in discrete points in the timeline.

3. Business Process Analysis Model

3.1 Simulation-based Performance Evaluation

Simulation can be accomplished by executing many various cases to simulate practical workflow instances. As to the performance evaluation of workflow models, following performance KPI are considered:

a) From the outside point of view, parameters used to measure performance of the whole model include: workflow case, average finishing time ($\overline{FT}$), average waiting time ($\overline{WT}$), average processing time ($\overline{PT}$);

$$\overline{FT} = \overline{WT} + \overline{PT}$$

Calculation formulas about $\overline{FT}$ $\overline{WT}$ and $\overline{PT}$ are listed as follows: (cc means case count)

$$\overline{FT} = \frac{\sum_{t=1}^{cc} FT_t}{cc}$$

$$\overline{WT} = \frac{\sum_{t=1}^{cc} WT_t}{cc}$$

$$\overline{PT} = \frac{\sum_{t=1}^{cc} PT_t}{cc}$$

b) From the internal point of view, parameters used to measure performance of service units are as follows: utilization of service units (μ) is equal to the case intensity（CI）divided by processing capability (PC), and service time (st) stands for total service time of service unites.

$$\mu = \frac{CI}{PC} = \frac{\sum_{t=1}^{cc} PT_t}{r \cdot st}$$

3.2 Error Analysis

Workflow scheduling and mutual exclusion of the resource semaphore should be processed during the course of collecting original data. Time data errors are caused because of additional overhead. Error processing is introduced as follows.

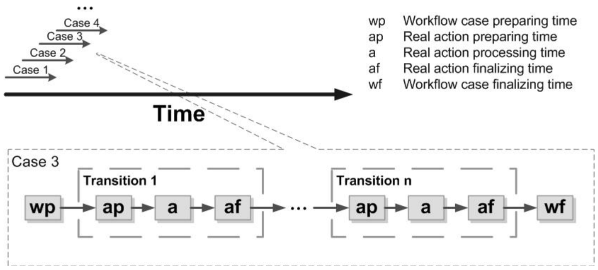

Figure 1. Error Source

Preprocessing of cases, which would consume the time of wp, should be accomplished first during the operation of a case simulation. There are preparing time (ap) and real action finalizing time (af) in the beginning and the end, as well as workflow case finalizing time (wf) before the end of a case.

The exclusion of wp and wf from the finishing time of cases would lead to the rising of cases' arrival time interval, witch is equal to wp+wf, and change the workflow model. Therefore wp and wf can only be calculated within the finishing time of cases. For workflow cases, relative error for finishing time RE_{FT} is calculated as follows: (tc stands for transition count of the specific case being served)

$$RE_{FT} = \frac{wp + wf + (ap + af) \cdot tc}{a \cdot tc}$$

The formula of average relative error $\overline{RE_{FT}}$ for finishing time is given as follows: (cc stands for the case count)

$$\overline{RE_{FT}} = \frac{\sum_{i=1}^{cc} (\mathrm{RE}_{FT})_i}{cc}$$

On the other hand, considering the situation of transition, the exclusion of ap and af from the processing time of transition means that there would be a certain idle time before and after the processing of a transition when a case is ready, and on the aspect of the case, it should be processed in this transition in a minute while nothing happens during the time interval of ap+af (there is only the idle time of ap before the first transition), which doesn't match the description of the workflow model neither. Therefore processing time error of cases is generated by calculating ap and af with real processing time in the processing time of the transition. The processing time is increased by (ap+af) multiplied by the number of transitions. The formula of relative error for processing time RE_{PT} is:

$$RE_{PT} = \frac{(ap + af) \cdot tc}{a \cdot tc}$$

The calculation of average relative error $\overline{RE_{FT}}$ for processing time is:

$$\overline{RE_{PT}} = \frac{\sum_{i=1}^{cc} (RE_{PT})_i}{cc}$$

Considering WT is actually equal to the value of waiting time error, relative error and average relative error are all equal to zero.

4. System Analysis and Design

4.1. The Goal of the BPSAS

The goal of workflow based BPSAS is to provide a platform which could analysis business processes and achieve following functions with given process models (including workflow models and resource allocation conditions).

a) Draw workflow model graph which reflects the model of workflow and provides basis for the conversion of model files which are supported by every workflow engine.

b) Analysis the performance of workflow model and reflect characters of processes by different performance indicators to provide basis for decision-makers to select the most suitable process model.

The overall structure of this system is shown as below:

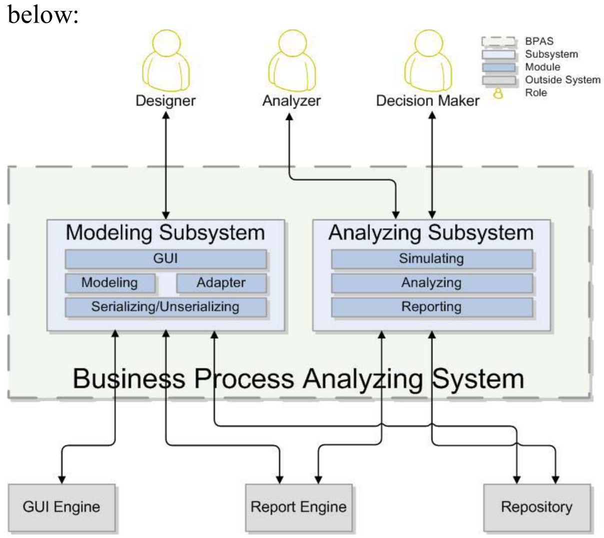

Figure 2. The logical structure of BPSAS system

4.2. Case Analysis

Based on the Application Data Center (ADC) business platform of this operation business company, we model the business processes for SI business, and the processes are described as follows:

1. First trial. The provincial company makes first trial to SI and relating local industry applications. Processing resources needs 3 hours and the time overhead is 8 hours.
2. Testing. The provincial company makes network access testing after the first trial. Processing resources needs 3 hours and the time overhead is 24 hours.
3. Configuration. Managers of this operation business company login on the ADC platform, select an application that needs to be added, define the name, description, tariff, preferential ratio of this application. Processing resources needs 3 hours and the time overhead is 4 hours.
4. Deployment. The network department and business supporting systems are entrusted to process the bureau data. Processing resources needs 2 hours and the time overhead is 8 hours.
5. Fulfillment. Local industry application is fulfilled in this step. Processing resources needs 1 hours and the time overhead is 2 hours.

Besides, the number of cases adds up to 2 everyday, (8 hours a day, intervals for 4 hours), a total of 50 cases need to be processed. The business process performance is evaluated under the condition of every case being processed successfully.

4.3. Experimental Design

The configuration and deployment are parallel in the model below after the first trail and testing, and accomplish the fulfillment step finally. The business model No.1 is shown as below.

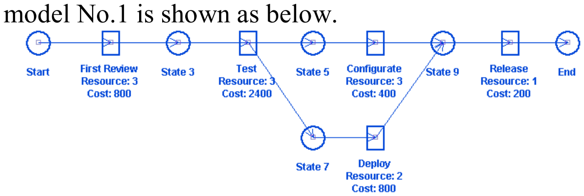

Figure 3. The experiment process model

Simulation situations of cases, internal and external performance KPI of this model are listed as follows:

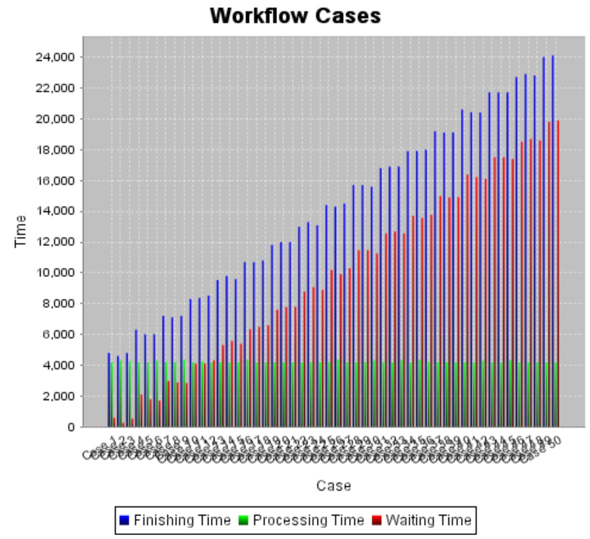

Figure 4. Case simulation result of model-1

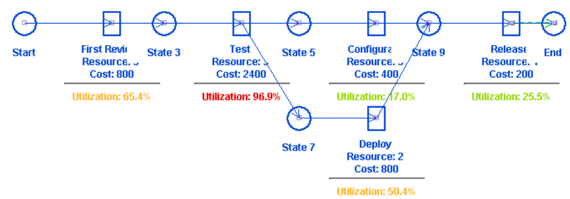

Figure 5. Internal performance indicators of model-1

Table 1. External performance indicators of model1

AFT	AWT	APT
142.24 （REAFT =0.4%）	99.83 (REAFT =0.0%）	42.40 (REAFT =1.0%）

We can see that the average finishing time and average service time are declining because the parallelism of configuration and deployment. As to the transition utilization, the transition utilization with less resources and high cost is relative greater than the transition utilization with more resources and low cost, which is accord with the general situation of low utilization of resources. From two parallel transitions, we can see the transition utilization with scarce resources and high consumption is relative high, which indicates resources in this transition are comparative busy.

4.4. Process Improvement

For example, we can make some adjustment about the model above as model No.2, in which we move two resources in relative idle transition configuration to relative busy transition testing on the premise that resources can do testing. The model is shown as below:

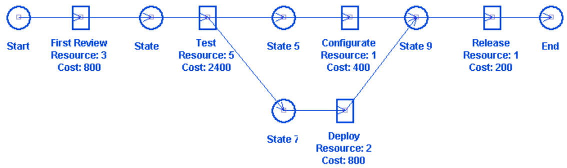

Figure 6. The improved model

The analysis result is as follows:

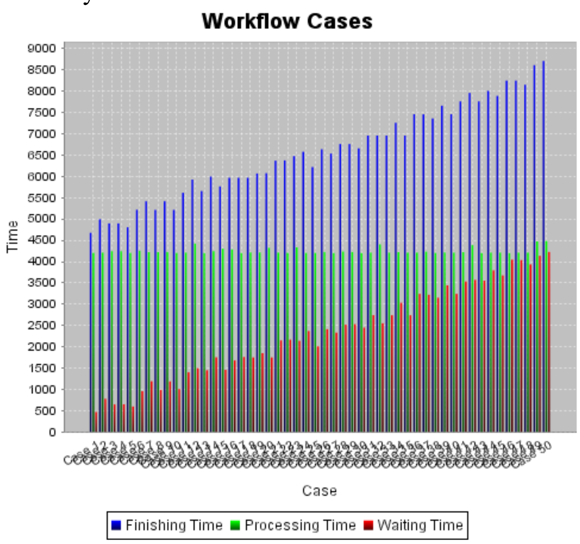

Figure 7. Case simulation result of model-2

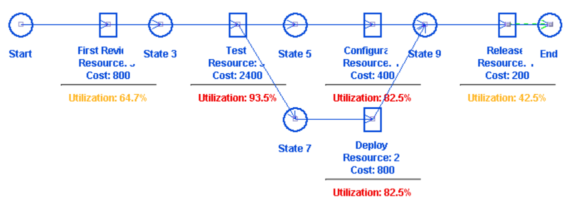

Figure 8. Internal performance indicators of model-2

Table 2. External performance indicators of model-2

AFT	AWT	APT
65.80 （REAFT =0.7%）	23.32 （REAFT=0.0% ）	42.48 (REAFT =1.0%）

From figures above, we can see that average finishing time and average waiting time are shortened in the external performance indicators while the busy degrees of transition testing, configuration and deployment tend to be uniform, which means their resources are allocated appropriately and they are all in efficient working conditions. Utilization of transition testing is relative high because the frequency of case arrival lets the processing of transitions in relatively full working state.

5. Conclusions

A business process simulation and analysis system, which is separated from specific workflow engines, is built based on the analysis and evaluation system of business process performance. Improved BPM procedure is supported by this system. The risk of unreasonable model design can be avoided to some extent.

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